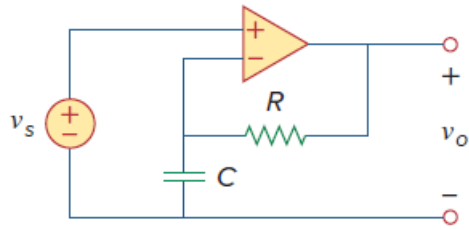


Problem

Calculate the gain $H(s) = V_o/V_s$ in the op amp circuit 4 of Fig. 16.100.

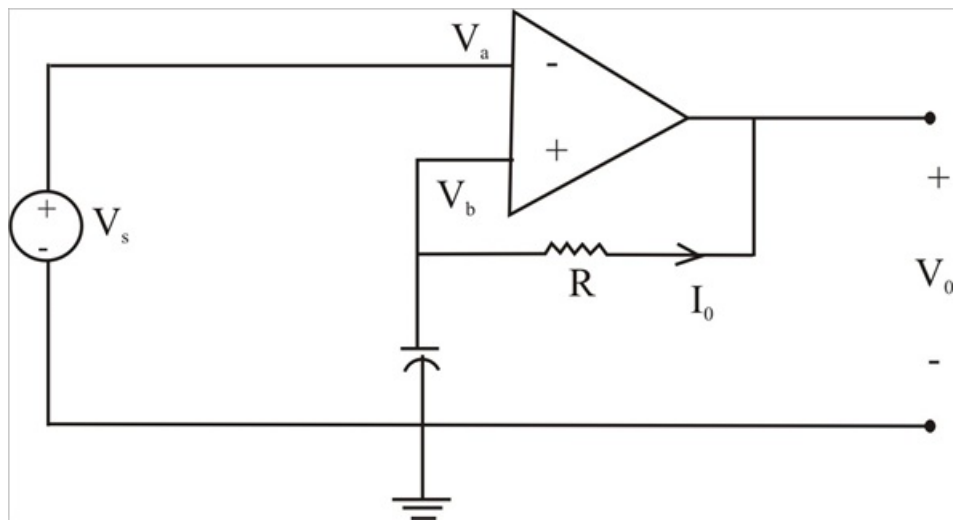
Figure 16.100:



Step-by-step solution

Step 1 of 2

Consider the circuit below



Step 2 of 2

Since no current enters the Op Amp, I_o flows through both R and C

$$V_o = -I_o \left(R + \frac{1}{sC} \right)$$

$$V_a = V_b = V_s = \frac{-I_o}{sC}$$

$$H(s) = \frac{V_o}{V_s} = \frac{R + \frac{1}{sC}}{\frac{1}{sC}} = sRC + 1$$